

AI ENGINEER

Mahmoud Ahmed Farouk

*AI Engineer | Machine Learning & Deep Learning | Computer Vision | Automation | Python*Cairo, Egypt | maf.bns@gmail.com | [LinkedIn](#) | [GitHub](#)

Professional Summary

AI Engineer focused on machine learning, deep learning, computer vision, retrieval systems, and automation. Software-development experience supports turning model work into interfaces, APIs, and complete human-facing products.

Core Expertise

- **Machine learning:** scikit-learn, XGBoost, LightGBM, Pandas, NumPy, structured experiments, and evaluation.
- **Deep learning and vision:** TensorFlow, Keras, OpenCV, CNNs, transfer learning, and DCGAN workflows.
- **Retrieval and automation:** ChromaDB, embeddings, Gemini, grounded RAG, source attribution, and Python automation.
- **Product engineering:** FastAPI, Django REST, Flutter, PySide6, PostgreSQL, SQLAlchemy, and SQLite.

Experience & Training

Project Management & Admin Support | Eram Group

January 2025 - Present

- Coordinate project tasks, monitor progress, prepare reports, support team communication, and assist daily operations.

AI & Machine Learning Trainee | INSTANT Software Solutions

July 2025 - December 2025

- Completed practical work across machine learning, TensorFlow, CNNs, RNNs, computer vision, NLP, YOLO, OpenCV, and introductory LLM workflows.

AI & Machine Learning Trainee | Div Academy

August 2024 - October 2024

- Completed practical machine-learning training recorded in the verified source CV.

CCNA Trainee | IT Gate Academy

August 2023 - October 2023

- Studied networking fundamentals, routing, switching, and troubleshooting.

IT Team Member | Enactus

January 2023 - May 2023

- Participated in the technical student team and supported its recorded IT activities.

Education

Computer Science Graduate | Modern Academy

2022 - 2026 | Class of 2026

- Graduated in Computer Science, Class of 2026.

Selected Projects

GenoScene | TEAM GRADUATION PROJECT

- Contributed to a Flutter, Django REST, FastAPI, PostgreSQL, and tabular-ML workflow that reports probabilistic eye, hair, and skin traits. Results are research-oriented, and any portrait is illustrative.

OralVision | COMPUTER VISION RESEARCH

- Built the documented six-class comparison and FastAPI interface. EfficientNet-B3 achieved repository-reported 98.41% accuracy and 98.14% macro F1. This classifier is not a clinical diagnosis service.

Corrective RAG System | RETRIEVAL AND AUTOMATION

- Implemented ingestion for PDF, DOCX, TXT, and CSV; local embeddings and ChromaDB retrieval; Gemini relevance evaluation; one bounded query correction; and visible source attribution.

Realistic Face DCGAN | GENERATIVE DEEP LEARNING

- Implemented and trained a TensorFlow/Keras DCGAN on 30,000 CelebA images for 30 epochs, producing synthetic 64 x 64 face grids and documented final training losses.

Point of Sale System | DESKTOP APPLICATION

- Built documented PySide6, SQLAlchemy, and SQLite workflows for roles, inventory, payments, refunds, loyalty, reporting, and kitchen operations.

GenoScene Web Prototype | BILINGUAL PROTOTYPE

- Developed an earlier JavaScript, Python, HTML, and CSS prototype that explored the GenoScene interaction and information design before the mobile product matured.

Certificates

- AI Diploma | INSTANT Software Solutions | 170 hours | 1 December 2025
- Digitopia 2025 | MCIT Egypt | Software Solutions and AI track
- Build with AI - Masr Edition | Google for Developers & ITI | 2025